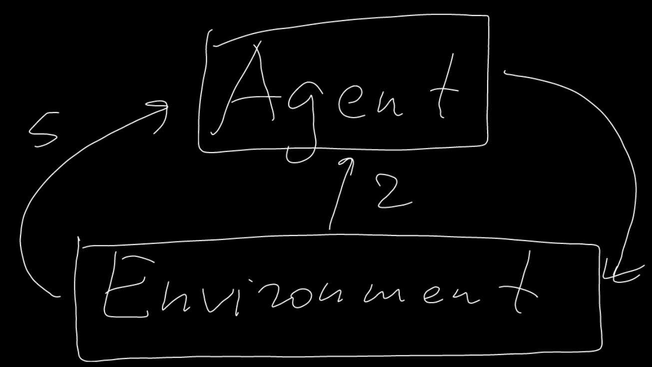
NNs for control: Reinforcement Learning

ML/DL Dataset $\{x_i, y_i\}_{i=1}^{N}$, $f(x, \theta)$

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{i=1}^{N} L(y_i, f(x_i, \theta)) + \lambda R(\theta) \to \min_{\theta}$$

$$\theta = \text{opt\_step}(\nabla_\theta \mathcal{L})$$

RL



$s \in S$ – state

$a \in A$ – action

$\pi(a|s)$ – policy

$p(s'|s,a)$ – transition probability

$r(s,a)$ – reward

Trajectory

$$\tau = \{ s_0, a_0, r_0, s_1, a_1, r_1, \ldots \}$$

$$p(\tau|\pi) = p(s_0)\,\pi(a_0|s_0)\,p(s_1|s_0,a_0)\,\pi(a_1|s_1)\ldots$$

Markov Decision Process (MDP)

$$p(s_t|s_{t-1}, a_{t-1}, s_{t-2}, a_{t-2}, \ldots, s_0, a_0) = p(s_t|s_{t-1}, a_{t-1})$$

$$R_t := r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots \quad , \quad \gamma \in (0,1)$$

discounted cumulative reward / return

$$J(\pi) = \mathbb{E}_{\tau|\pi} R_0 \longrightarrow \max_{\pi}$$

$$V^{\pi}(s) := \mathbb{E}[R_t \mid s_t = s]$$ – value function
state function

$$V^{*}(s) := \max_{\pi} V^{\pi}(s)$$

$$Q^{\pi}(s,a) := \mathbb{E}[R_t \mid s_t = s, a_t = a]$$ – quality function
state-action func.

$$Q^{*}(s,a) = \max_{\pi} Q^{\pi}(s,a)$$

$$R_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots = r_t + \gamma\left(r_{t+1} + \gamma r_{t+2} + \dots\right) =$$
$$= r_t + \gamma R_{t+1}$$

$$V^{\pi}(s_0) = \mathbb{E}_{a_0, s_1, a_1, \dots} R_0 = \mathbb{E}_{a_0, s_1, a_1, \dots}\left(r(s_0, a_0) + \gamma R_1\right) =$$

$$= \mathbb{E}_{\pi(a_0|s_0)}\left(r(s_0, a_0) + \gamma \mathbb{E}_{p(s_1|s_0,a_0)} \mathbb{E}_{a_1, s_2, a_2 \dots} R_1\right) \Rightarrow$$

$$\Rightarrow \boxed{V^{\pi}(s) = \mathbb{E}_{\pi(a|s)}\left(r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} V^{\pi}(s')\right) \quad \forall s} \quad V^{\pi}(s_1)$$

$$V^*(s_0) = \max_{\pi_0(a_0|s_0),\, \pi_1(a_1|s_1),\, \dots} V^\pi(s_0) =$$

$$= \max_{\pi_0, \pi_1, \dots} \mathbb{E}_{\pi_0(a_0|s_0)} \left(r(s_0, a_0) + \gamma \mathbb{E}_{p(s_1|s_0, a_0)} V^\pi(s_1) \right) =$$

$$= \left\{ \max(a + b, a + c) = a + \max(b, c) \right\} =$$

$$= \max_{\pi_0(a_0|s_0)} \mathbb{E}_{\pi_0(a_0|s_0)} \left(\underbrace{r(s_0, a_0) + \gamma \mathbb{E}_{p(s_1|s_0, a_0)} \underbrace{\max_{\pi_1, \pi_2, \dots} V^\pi(s_1)}_{V^*(s_1)}}_{Q^*(s_0, a_0)} \right)$$

$$\mathbb{E}_{\pi(a|s)}\, y(s,a) = \begin{cases} \sum_a \pi(a|s)\, y(s,a) \to \max_\pi \\ \sum_a \pi(a|s) = 1, \quad \pi(a|s) \geq 0 \ \forall a \end{cases}$$

$$\Leftarrow$$

$$\pi_{opt}(a|s) = \mathbb{I}\left(a = \arg\max_a y(a,s)\right)$$

$$\boxed{V^*(s) = \max_a \left(r(s,a) + \gamma\, \mathbb{E}_{p(s'|s,a)} V^*(s') \right)}$$

$$\pi_{opt}(a|s) = \arg\max_a \left(\ldots \right)$$

$$Q^{\pi}(s,a) = r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} \mathbb{E}_{\pi(a'|s')} Q^{\pi}(s',a')$$

$$Q^{*}(s,a) = r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} \max_{a'} Q^{*}(s',a')$$

$$V^{\pi}(s) = \mathbb{E}_{\pi(a|s)} Q^{\pi}(s,a)$$

$$V^{*}(s) = \max_{a} Q^{*}(s,a)$$

$$Q^{\pi}(s,a) = r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} V^{\pi}(s')$$

$$Q^{*}(s,a) = r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} V^{*}(s')$$

Value Iteration

Init. $V(s)\ \forall s$

For iterations:

$\Delta = 0$

For all states s:

$\delta = V(s)$

$$V(s) = \max_a \left(r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} V(s') \right)$$

$$\Delta = \max(\Delta, |V(s) - \delta|)$$

if $\Delta \leq \varepsilon$, then stop

Inference:

$$\pi(a|s) = \arg\max_a \left(r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} V(s') \right)$$

Example

terminal states

s – position

$a \in \{\leftarrow, \rightarrow, \uparrow, \downarrow\}$

reward:

0	0	0	1
0		0	−1
0	0	0	0

$\max(0 + \gamma(-1),\ 0 + \gamma \cdot 0) = 0$

V_0:

0	0	0	1
0		0	−1
0	0	0	0

$\rightarrow V_1$:

0	0	γ	1
0		γ^2	−1
0	0	0	0

$\rightarrow V_2$:

γ^7	γ^2	γ	1
γ^6		γ^2	−1
γ^5	γ^4	γ^3	0

$\rightarrow$

$\rightarrow V_3$:

γ^3	γ^2	γ	1
γ^6		γ^2	−1
γ^5	γ^4	γ^3	γ^4

$\rightarrow V_4 = V_*$:

γ^3	γ^2	γ	1
γ^4		γ^2	−1
γ^5	γ^4	γ^3	γ^4

Q-learning

$$\forall s, a$$

$$Q^*(s,a) = r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} \max_{a'} Q^*(s',a')$$

$$\Downarrow$$

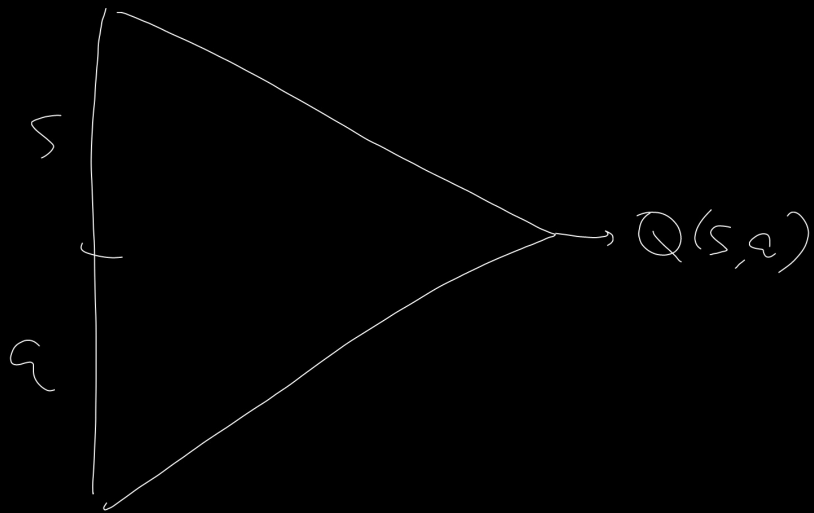
$$L(Q) = \mathbb{E}_{s,a} \left\{ \frac{1}{2} \left(Q(s,a) - \left(r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} \max_{a'} Q(s',a') \right) \right)^2 \right\} \to \min_Q$$

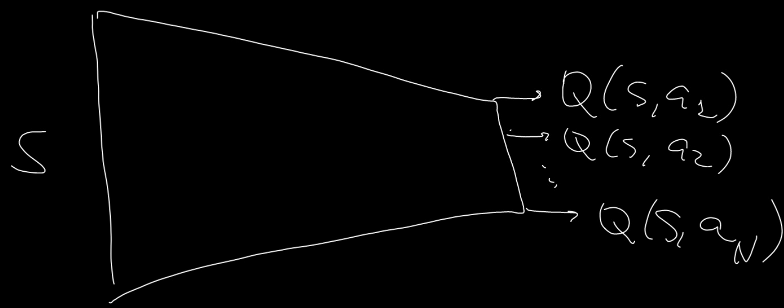
$$\sim s, a, r, s', \quad y(s,a) = r + \gamma \max_{a'} Q(s',a')$$

$$\text{SGD: } Q_{new}(s,a) = Q_{old}(s,a) - \alpha \left(Q_{old}(s,a) - y(s,a) \right)$$

$$Q(s, a | \theta)$$

$$L(\theta) = \mathbb{E}_{s,a} \frac{1}{2} \left(Q(s, a | \theta) - y(s, a) \right)^2 \to \min_{\theta}$$





DQN (Deep Q Network)

Init. θ, experience replay buffer $M = \emptyset$

For episodes:

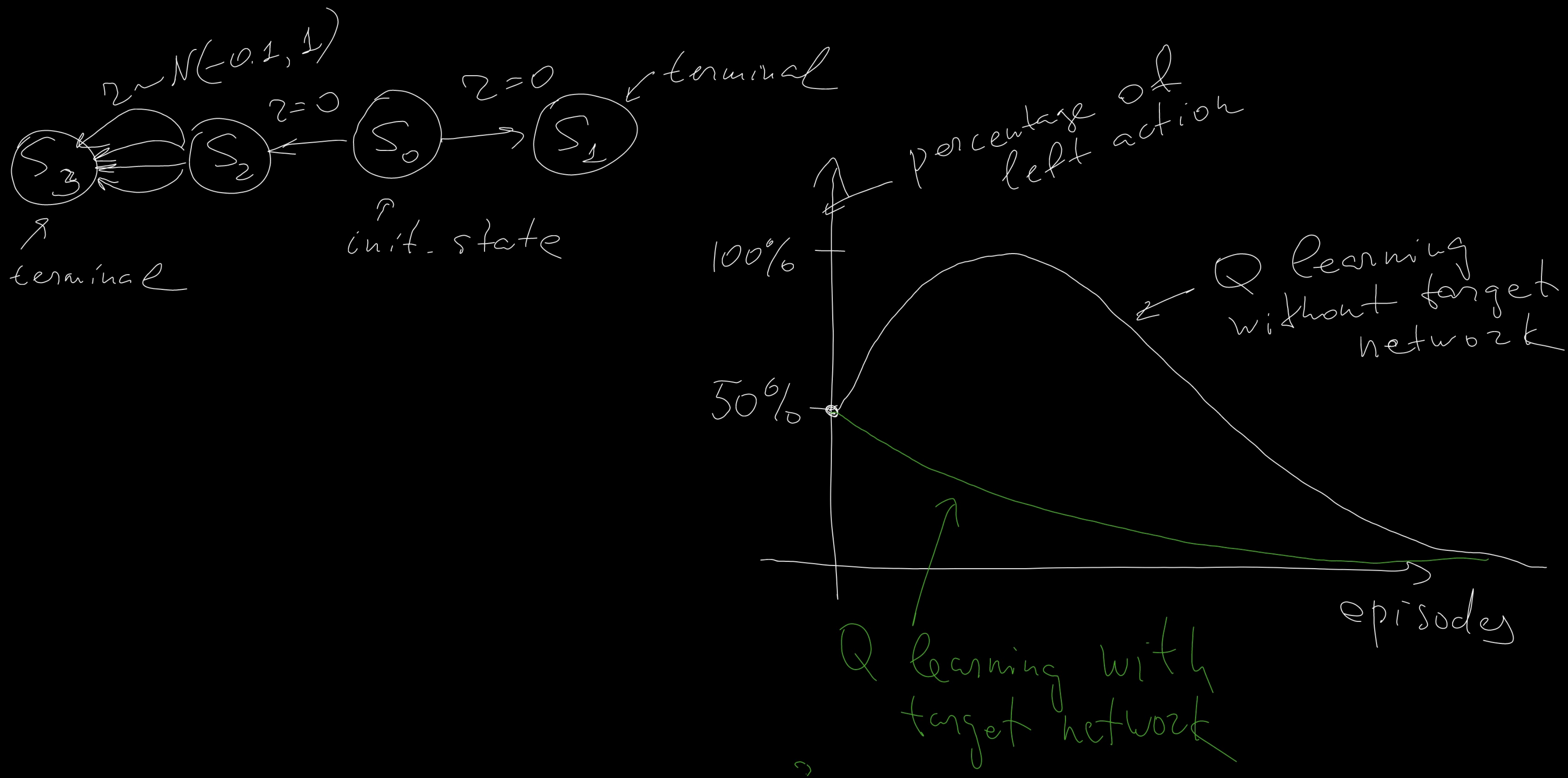
- Init. s_0
- For $t = 0, 1, 2, \dots, T$:
  - $a_t \sim \varepsilon\text{-greedy}(Q(s_t, a))$
  - get r_t, s_{t+1}, $(s_t, a_t, r_t, s_{t+1}) \longrightarrow M$
  - sample mini-batch $\{(s_i, a_i, r_i, s'_i)\}_{i=1}^{B}$ from M
  - for $i = 1, \dots, B$:
    - $y_i = r_i + \gamma Q\left(s'_i, \arg\max_a Q(s'_i, a \mid \theta) \mid \bar{\theta}\right)$
    - $\mathcal{L}_\theta = \mathcal{L}_\theta + \frac{1}{B}\left(Q(s_i, a_i \mid \theta) - y_i\right)^2$
  - $\theta = \text{opt\_step}(\nabla_\theta \mathcal{L})$

$$\varepsilon\text{-greedy}: \begin{cases} \text{random action with prob. } \dfrac{\varepsilon}{|A|-1} \\ \arg\max_a Q(s,a), \text{ with prob. } 1-\varepsilon \end{cases}$$

$\bar{\theta}$ – target network

detach

$$\bar{\theta} = (1-\tau)\bar{\theta} + \tau\theta$$



r~N(-0.1, 1)
r=0
r=0
terminal
S3
S2
S0
S1
init. state
terminal
percentage of left action
100%
50%
Q learning without target network
episodes
Q learning with target network